

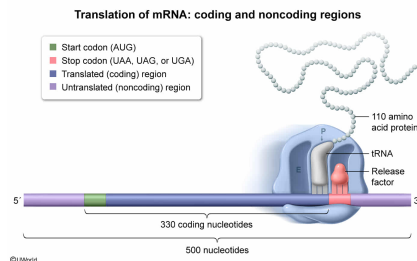
Following intron splicing in the nucleus, a mature mRNA molecule consists of 500 nucleotides and is translated into a 110 amino acid protein. Given this, which of the following statements regarding this mRNA molecule would be true?

- ☐ A. 110 nucleotides of the mRNA are translated into amino acids of the protein. [5%]
- ☐ B. The mRNA was transcribed from a 500 base-pair gene. [11%]
- ✓ ☒ C. 170 nucleotides of the mRNA are not translated into amino acids of the protein. [73%]
- ☐ D. The protein translated from the mRNA has a molecular weight of approximately 5.5 kDa. [9%]

Correct

73% answered correctly

Explanation:



In **eukaryotic cells**, nuclear genes are initially **transcribed** by RNA polymerase to **pre-mRNA**. Pre-mRNA is then **converted** into mature mRNA in a process that includes **splicing to remove introns**. Mature mRNA is then transported out of the nucleus into the cytoplasm, where it can be **translated** into protein by ribosomes.

Each amino acid in a protein is encoded by a corresponding mRNA **codon**, a sequence of *three* consecutive nucleotides that binds the **anticodon** of a tRNA "charged" with a specific amino acid. Translation of mRNA (in the 5' to 3' direction) initiates at the **start codon** (AUG) and continues until a **stop codon** (UAA, UAG, or UGA) is reached. The stop codon does *not* code for an amino acid but instead binds a release factor that promotes dissociation of the translation complex. In addition, regions *upstream* of the start codon and *downstream* of the stop codon—referred to as **untranslated regions (UTRs)**—do *not* code for amino acids.

In this scenario, a mature mRNA molecule consisting of 500 nucleotides was translated into a 110 amino acid protein. Because each amino acid is encoded by a three-nucleotide codon, the number of nucleotides translated into this protein can be calculated:

$$110 \text{ amino acids} \times 3 \text{ nucleotides per codon} = 330 \text{ nucleotides}$$

Therefore, the *330 nucleotides* of this mRNA molecule are translated into amino acids of the protein (**Choice A**). The number of nucleotides that do *not* code for amino acids can then be calculated:

$$500 \text{ total nucleotides} - 330 \text{ coding nucleotides} = 170 \text{ noncoding nucleotides}$$

Therefore, **170 nucleotides** of this mRNA molecule, including the stop codon and UTRs, **are not translated into amino acids** of the protein.

(Choice B) The question states that the mature mRNA molecule contains 500 nucleotides *after* intron splicing. Therefore, the original pre-mRNA molecule was most likely transcribed from a gene that contained *more* than 500 nucleotides.

(Choice D) The average weight of an amino acid is 110 Da. Therefore, the approximate weight of a 110 amino acid protein is about 12,100 Da, or 12.1 kDa (**1,000 Da = 1 kDa**).

Educational objective:

Each amino acid in a protein is encoded by a corresponding three-nucleotide mRNA codon. However, mRNA molecules also include regions that are not translated into the protein (ie, a stop codon and untranslated regions).

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